



Measuring & Energy

Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 Write 2 871 501 in standard form correct to 2 significant figures. 1

2 871 501 rounded to 2 significant figures is 2 900 000 which is 2.9×10^6

- 2 The height of a building is measured. It is found to be 480 m, to the nearest 10 m. What are the lower and upper limits of the true measurement? 1

$\pm 5m$

475 m and 485 m.

- 3 The sides of a rectangle have been measured to be 3.6 cm and 1.3 cm. a. Calculate the percentage error for the measurement of the longest side. 1

Accurate to nearest 0.1 cm.

$$\text{Absolute error} = \pm 0.5 \times 0.1 = \pm 0.05$$

$$\text{Percentage error} = \frac{0.05}{3.6} \times 100\% = 1.39\% \text{ (2 dec places)}$$

- b. The area of the rectangle has been calculated to be 4.68 square centimetres, using the measurements stated above. Express this as a percentage of the largest possible area 2

for this rectangle, considering the absolute error for the measurements.

Calculate the area using the upper limits of the 2 measurements

$$3.6 + 0.05 \text{ and } 1.3 + 0.05 = 3.65 \text{ and } 1.35$$

$$\text{maximum area} = 3.65 \times 1.35 = 4.9275 \text{ cm}^2$$

Express 4.68 as a percentage of this maximum area:

$$\frac{4.68}{4.9275} \times 100\% = 94.9771895 \approx 95\%$$

- 4 What is 54.9864 rounded to 3 significant figures? 1

55.0

- 5 The table shows the average energy used in kilojoules per kilogram of body mass, when cycling at two 2

different speeds for 30 minutes. Shane weighs 62 kg.

Cycle Speed	Average energy used (kJ/kg) in 30 minutes
15 km/h	11.52
20 km/h	18.43

If 1 calorie is equivalent to 4.184 kilojoules, approximately how many more calories would

Shane use, cycling at 20 km/h, than he would at 15 km/h during the 30 minutes? Answer

correct to the nearest whole number.

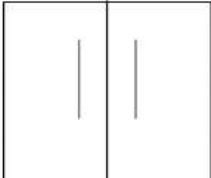
$$(18.43 - 11.52) \times 62 \div 4.184 = 102.3948 \dots \approx 102$$

- 6 A pencil is measured to be 13.6 cm long. What is the percentage error of this measurement, correct to two decimal places? 1

$$\text{absolute error} = \frac{1}{2} \times \text{precision} = 0.5 \times 0.1 = 0.05 \text{ cm}$$

$$\% \text{ error} = \frac{\text{absolute error}}{\text{measurement}} \times 100 = \frac{0.05}{13.6} \times 100 = 0.37\%$$

- 7 Nikki needs to buy a new refrigerator. She has found two brands with all the features she requires. Their prices and power consumption are shown here. The average cost for electricity is 26c/kWh. Nikki's fridge will be switched on 24 hours per day, every day.

Option A	Option B
	
Purchase price \$4399	Purchase price \$3098
Power consumption 64 W	Power consumption 57 W

- a. If Nikki chooses Option A, her annual electricity costs for the refrigerator will be \$145.77. Calculate Nikki's annual electricity costs if she chooses Option B.

Option B electricity costs:
 $57 \times 24 \times 365 = 499\,320 \text{ kWh per year}$
 $499\,320 \div 1000 \times 0.26 = \$129.8232 \approx \$129.82 \text{ per year}$

- b. Nikki has been offered the following finance deals for the two refrigerators:

Option A	Option B
Monthly repayment = \$102.27	Monthly repayment = \$78.59
Term = 4 years	Term = 5 years

Assuming electricity costs remain the same, the total cost to purchase Option A over the 4 year finance term, including electricity costs is \$5492.04. Calculate the total costs to purchase Option B.

Option B total costs over 5 years:
 5 years = 60 months
 $\text{Cost} = 78.59 \times 60 + 5 \times 129.82 = \5364.50

- c. The cost of Option A over a 5 year period (to allow comparison), would include one more year of electricity costs as shown here:

$$\$5492.04 + \$145.77 = \$5637.81$$

How much would the monthly repayments for Option A need to decrease by so that its total 5-year cost is the same as for Option B?

Costs over 5 years:
 Option A = \$5637.81
 Option B = \$5364.50
 Difference = \$273.31
 Option A repayments are over 4 years (48 months)
 Monthly repayments would need to decrease by:
 $273.31 \div 48 = 5.693958333 \approx \5.69

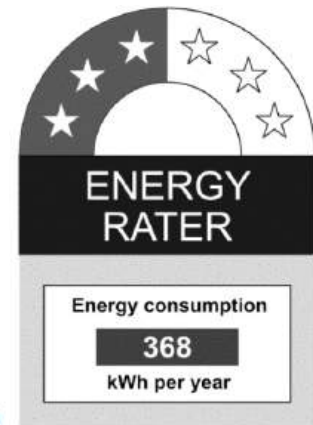
- 8 Samuel measured his height to be 173 cm, correct to the nearest centimetre. What is the percentage error in his measurement?

$$\pm 0.29\%$$

- 9 A 2500-watt vacuum cleaner has the following energy label on it.

3

The energy label assumes that a typical household would consume 368 kWh of energy per year using the vacuum cleaner. How many hours and minutes of usage would this equate to each week? Give your answer correct to the nearest minute. Assume 52 weeks per year.



$$\begin{aligned} \text{Assumed number of kWh per week} &= \frac{368}{52} = 7.076... \text{ kWh} \\ \text{energy rating of vacuum} &= 2500 \text{ W} = 2.5 \text{ kW} \\ \text{Power consumption} &= \text{power rating} \times \text{time} \\ 7.076... &= 2.5 \times t \\ t &= \frac{7.076...}{2.5} = 2.8307... \text{ h} \\ &= 2 \text{ h } 49 \text{ min } 50.77 \text{ sec.} \\ &= 2 \text{ hours } 50 \text{ min (nearest min)} \end{aligned}$$

Correct rounding

- 10 The cost of gas is determined using the table below:

Usage Charge	First 3000 MJ	\$0.05280 per MJ
	Additional MJ over 3000	\$ 0.04620 per MJ

- a. Maria used 7000MJ of gas in this period. What is the charge?

2

$$3000 \times 0.05280 + 4000 \times 0.04620 = 343.20$$

- b. Maria has decided to reduce her energy bills. She has a target of \$200 for gas. What is the maximum number of MJ she is allowed in this period? Answer correct to the nearest megajoule.

3

$$\begin{aligned} 3000 \times 0.05280 &= 158.40 \\ 158.40 + x \times 0.04620 &= 200 \\ x &= \frac{41.60}{0.04620} = 900 \text{ MJ} \\ \text{max usage} &= 3900 \text{ MJ} \end{aligned}$$

- 11 What is 7 400 000 000 written in standard form?

1

$$7.4 \times 10^9$$

- 12 Convert 27 500 000 kW to MW.

1

$$\frac{27\,500\,000}{1\,000\,000} \times 1000 = 27\,500 \text{ MW}$$

13 Sam has a household fridge that uses energy at a rate of 738 kWh per year. She is thinking of buying a bar fridge to store extra bottles of wine. The cost of the bar fridge is \$420 and it has an energy consumption rate of 313 kWh per year.

- a. If energy is charged at the rate of \$0.32/kWh, what is the energy cost of running the bar fridge for one year? 2

$$0.32(313) = \$100.16$$

- b. What would be the percentage increase in energy costs to run both fridges compared to running only the household fridge? Give your answer correct to the nearest percent. 2

$$\begin{aligned} 0.32(738) &= \$236.16 \\ \text{Both fridges} &= \$336.32 \\ \frac{336.32 - 236.16}{236.16} \times 100\% &\doteq 42\% \end{aligned}$$

- c. What would be the total cost of buying the bar fridge and running it for five years? 2

$$420 + 5(100.16) = \$920.80$$



Measuring & Energy Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 What speed is equivalent to 70 km/h? $\frac{70 \times 1000}{60^2}$ 19.4 m/s 1
- 2 2.3×10^{-4} is the same as: 1
- A. 0.23×0.01 B. 0.023×0.01
- C. 0.00023×0.1 D. 23000
- 3 For lunch, Mark has a chicken and avocado wrap with an energy content of 1972 kJ. When he rides his bike, he burns 1850 kJ energy per hour. How many minutes must he ride his bike to work off the energy from the chicken and avocado wrap. $1972/1850 \times 60 = 64$ mins 1
- 4 a. Lance has a solar panel system installed in his house. How much power will this generate in 365 days if the average daily production of the solar panel system is 7.25 kWh? $7.25 \times 365 = 2\,646.25$ kWh 1
- b. Lance's average annual household consumption is 9856 kWh. What percentage of the household consumption is generated by the solar panel system? Answer correct to one decimal place. % generated by solar panel = $2\,646.25/9\,856 \times 100\% = 26.8\%$ 1
- c. Lance also wants to know how much it costs to charge his laptop each year. He uses a 120 W charger for his laptop and it takes about 30 minutes. He does this five times a week, every week of the year. If the cost of electricity is \$0.456725/kWh, calculate the cost of charging his laptop for six months of the year. 2
- Laptop power used for 6 months = $120\text{ W} \times 30/60 \times 5 \times 26 = 7800\text{ Wh} = 7.8\text{ kWh}$
- Total cost for 6 months = $7.8 \times 0.456725 = \$3.56$
- 5 Express 0.004182 in standard form with two significant figures. 4.2×10^{-3} 1
- 6 The length of a beach is measured as 1.45 km. What is the absolute error of this measurement? $1.45\text{ km} = 1450\text{ m}$
smallest unit is 10 m
 $\pm \frac{10}{2} = 0.5\text{ m}$ 1
- 7 A flight between Sydney and the USA takes 14 hours, to the nearest hour. What is the percentage error in measurement, correct to 2 decimal places is? 3.57% 1

- 8 Last year the world population was 7 577 130 000. Write this number in scientific notation, 1

correct to four significant figures. 7.577×10^9

- 10 The length of the Sydney Harbour Bridge has been measured as 1149 m, correct to the nearest metre.

a. What is the precision of this measurement? Precision = 1 m 1

b. What is the absolute error of this measurement? Absolute error = 0.5 m 1

c. Calculate the percentage error for this measurement? Give your answer correct to 2 2

decimal places.
$$\text{Percentage error} = \frac{\text{Absolute error}}{\text{Measurement}} \times \frac{100}{1}$$
$$= \frac{0.5}{1149} \times \frac{100}{1} = 0.04351610096 = 0.04\%$$

- 11 The height of a door is measured to be 2.1 metres, correct to the nearest 10 centimetre. What is the absolute error of this measurement? $\pm 0.05 \text{ m}$ 1

- 12 Which of the following is equivalent to 2 000 μm ? 1

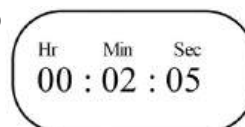
A. 0.02 m B. 20 cm C. 20 000 mm D. 2 000 000 nm

- 13 Alice uses 9 kilocalories of energy per minute while she is cycling. She consumes an energy drink that contains 895 kilojoules of energy. How many minutes will she need to cycle in order to use up all the energy from the energy drink, correct to the nearest minute? 2

(1 kilocalorie = 4.184 kilojoules)

$$\begin{aligned} 895 \div 4.1 &= 213.91 \text{ kJ} \\ 213.91 \div 9 &\approx 23.77 = 24 \text{ minutes (to the nearest minute)} \end{aligned}$$

- 14 Alex measures the number of seconds that it takes his pulse rate to return to exercising. The time taken is shown below on the timer app that he has on 1



his phone. What is the percentage error in the measurement of this time?

The smallest unit used by this timer is the second, so the absolute error is 0.5 sec. 2 minutes and 5 seconds is 125 seconds.

$$\% \text{ error} = \frac{0.5}{125} \times 100 = 0.4\%$$

- 15 The table shows the average energy used, in 2

kilojoules per kilogram of body mass, by a

person bicycling for 20 minutes at different

Bicycling speed	Energy used in 20 minutes
15 km/h	11.52 kJ/kg
20 km/h	18.43 kJ/kg

speeds. Pamela, who weighs 51 kg, eats a Kit-Kat containing 115 kilocalories. For approximately how long does she need to cycle at 15 km/h to burn off the energy contained in the Kit-Kat?

(1 kilocalorie = 4.184 kilojoules) 21 minutes

- 16 Last year, a total of 4 432 700 000 people visited the cinemas in India. Write this number in scientific notation. 1

$$4.4327 \times 10^9$$

- 17 Calculate the cost of running a 400-watt television set for nine hours if the average peak rate for domestic electricity is \$0.15/kWh. 2

$$\text{Total electricity used} = 400 \times 9 = 3600 \text{ watt-hours or } 3.6 \text{ kWh}$$

$$\text{Cost of electricity used} = 3.6 \times 0.15 = \$0.54$$

- 18 Robert is curious about his favourite candy-coated chocolates Smart-M's. He reads on the label that they contain 520 kJ per 25 gram serving. Using a scale, he discovers that 35 Smart-M's weigh 25 grams.

- a. How many kilojoules are in each Smart-M? Round your answer to the nearest whole kJ. 1

$$520 \text{ kJ} \div 35 = 14.85 \dots \text{kJ} \approx 15 \text{ kJ per Smart-M}$$

- b. Robert uses the following formula he found online to calculate the number of kilocalories he burns from walking: 2

$$C = 0.035W + \frac{V^2}{H} \times 0.029W$$

where C is kilocalories per minute, W is weight in kilograms, V is speed in metres per second, and H is height in metres.

If Robert weighs 100 kilograms, is 1.25 metres tall, and typically walks at a speed of 1.4 metres per second, use the formula and the fact that one calorie is approximately 4.2 joules to calculate the number of kilojoules he burns per minute from walking. Round your answer to the nearest whole kJ.

$$C = 0.035(100) + \frac{1.4^2}{1.25} \times 0.029(100) = 8.0472 \text{ kcal per minute} \checkmark$$

$$8.0472 \times 4.2 = 33.79824 \text{ kJ} \approx 34 \text{ kJ per minute} \checkmark$$

- c. Given your calculations in parts (a) and (b), find how far Robert needs to walk in order to burn the kilojoules from one Smart-M, to the nearest metre. 3

$$15 \text{ kJ} \div 34 \frac{\text{kJ}}{\text{min}} = 0.4417 \dots \text{min} \checkmark$$

$$0.4417 \dots \times 60 = 26.47 \dots \text{sec} \checkmark$$

$$1.4 \frac{\text{m}}{\text{s}} \times 26.47 \text{ s} = 37.05 \dots \text{m} \approx 37 \text{ m} \checkmark$$

(an answer within 1 m may be obtained depending on rounding or use of exact values in calculations)